

QUANTUM COMPUTING FOR HOSPITAL APPOINTMENT SCHEDULING

LITERATURE SURVEY

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2. LITERATURE SURVEY

Hospital appointment scheduling is a complex optimization problem because it requires the coordination of patients, doctors, time slots, resources, and multiple scheduling constraints. Existing healthcare scheduling research has studied patient appointments, nurse rostering, operating-room scheduling, and other resource-allocation problems using mathematical optimization, heuristics, and metaheuristics. Recent research has also started exploring quantum computing and hybrid quantum-classical methods for these optimization tasks.

2.1 Healthcare Appointment Scheduling

A comprehensive review by Ala et al. (2022) examined appointment scheduling in healthcare services and discussed different problem formulations and solution approaches. The study shows that appointment scheduling is an important healthcare operations problem involving multiple constraints and objectives. This work provides the classical healthcare scheduling foundation for the proposed project, where appointment assignment must consider practical requirements while improving the overall efficiency of the scheduling process.

2.2 Healthcare Scheduling as an Optimization Problem

A broader review of healthcare scheduling by Al-Huson et al. (2021) covered patient admission, nurse scheduling, operating-room scheduling, surgery scheduling, and related healthcare scheduling problems. The review emphasizes that scheduling research is important for improving patient flow, reducing operational costs, and making better use of hospital resources. These findings support the formulation of hospital appointment scheduling as an optimization problem in which different constraints and objectives must be handled together.

2.3 Quantum Annealing for Nurse Scheduling

Ikeda, Nakamura, and Humble (2019) investigated the use of quantum annealing for the Nurse Scheduling Problem using a D-Wave 2000Q quantum annealer. The scheduling problem was converted into an Ising-type Hamiltonian with hard constraints. Their experiments showed that quantum annealing could recover satisfying schedules for the tested problems. They also found that reverse annealing could improve solution quality by refining returned solutions. This study demonstrates that quantum optimization can be applied to practical healthcare scheduling problems with constraints.

2.4 Quantum Formulation of Scheduling Problems

Research on encoding optimization problems for quantum computing has demonstrated how scheduling constraints can be represented using binary variables and cost functions. For nurse scheduling, variables can indicate whether a particular nurse is assigned to a particular shift, while penalty terms can represent constraints such as workload balance and consecutive working days. Such formulations are relevant to hospital appointment scheduling because patient-doctor-time-slot assignments can similarly be represented using binary decision variables and constraint penalties.

2.5 Quantum Optimization of Physician Scheduling

Larsson and Kanerot (2025) studied the Physician Scheduling Problem using the Quantum Approximate Optimization Algorithm (QAOA). Their work formulated physician scheduling as a Quadratic Unconstrained Binary Optimization (QUBO) problem and considered factors such as availability, preferences, fairness, and contractual workload. Classical solvers such as Gurobi and Z3 were used as benchmarks, while QAOA was evaluated using simulators and IBM quantum hardware. Their results indicate that classical methods currently provide strong feasible solutions, while QAOA shows potential despite present quantum hardware limitations. This study is closely related to the proposed project because both involve healthcare scheduling and QAOA-based optimization.

2.6 QAOA for Scheduling

Recent research on QAOA has focused on constrained scheduling problems. A 2026 study on a QAOA-based Quantum Time Interval Scheduler formulated scheduling with strict temporal constraints as a QUBO problem. The approach separated objective and constraint Hamiltonians and introduced quantum-circuit mechanisms for detecting time collisions. The study demonstrates how QAOA can be adapted to scheduling problems where tasks must be assigned to non-conflicting time intervals. This concept is directly relevant to hospital appointments, where overlapping doctor appointments and unsuitable time assignments must be avoided.

2.7 Recent Quantum-Assisted Healthcare Optimization

Waseem, Epiphaniou, and Maple (2026) proposed a Hybrid Quantum-Classical Architecture for healthcare decision-making. Their framework uses classical preprocessing together with short-depth QAOA on NISQ-era quantum systems. One of the evaluated applications was operating-room scheduling. The reported results suggest that quantum-assisted optimization can be explored for constrained healthcare decision-making, although practical deployment still depends on hardware capability, noise, scalability, and problem size. The study supports the hybrid architecture used in the proposed project.

2.8 Research Gap

The literature shows that quantum computing has been explored for nurse scheduling, physician scheduling, operating-room scheduling, and general constrained scheduling. However, direct quantum approaches specifically targeting patient appointment scheduling remain comparatively less developed. Existing studies also indicate that current quantum hardware limits problem size and can make classical optimization methods competitive. Therefore, a practical project can focus on a small or medium-sized hospital appointment scheduling model and use a hybrid quantum-classical workflow. The proposed project addresses this gap by applying quantum optimization concepts, particularly QAOA, to appointment scheduling while considering doctor availability, patient requirements, appointment duration, and scheduling conflicts.

2.9 Summary of Literature

Study	Area	Method	Key Contribution
Ala et al. (2022)	Healthcare appointment scheduling	Review of scheduling approaches	Identifies appointment scheduling models, objectives, and solution approaches.
Al-Huson et al. (2021)	Healthcare scheduling	Literature review	Covers patient, nurse, surgery and operating-room scheduling and resource use.
Ikeda et al. (2019)	Nurse scheduling	Quantum annealing	Demonstrates feasible nurse schedules and benefits of reverse annealing.

Study	Area	Method	Key Contribution
Larsson & Kanerot (2025)	Physician scheduling	QUBO + QAOA	Studies availability, fairness and preferences using quantum and classical solvers.
QTIS (2026)	Time-interval scheduling	QUBO + QAOA	Handles temporal constraints and time-collision detection.
Waseem et al. (2026)	Healthcare optimization	Hybrid quantum-classical QAOA	Applies short-depth QAOA to constrained healthcare decision-making.

2.10 Conclusion

The literature indicates a growing interest in applying quantum optimization to healthcare scheduling. Quantum annealing has already been investigated for nurse scheduling, while QAOA has been studied for physician scheduling and constrained scheduling problems. At the same time, healthcare scheduling reviews show that appointment and resource scheduling involve many practical constraints. Based on these findings, the proposed project combines classical data processing with quantum optimization to explore hospital appointment scheduling. The main focus is to reduce waiting time, avoid appointment clashes, and improve doctor and resource utilization while recognizing the current limitations of quantum hardware.

References

1. Ala, A., Chen, F., et al. (2022). Appointment Scheduling Problem in Complexity Systems of the Healthcare Services: A Comprehensive Review. *Journal of Healthcare Engineering*. DOI: 10.1155/2022/5819813.
2. Al-Huson, M., et al. (2021). Healthcare scheduling in optimization context: a review. *BMC Health Services Research*.
3. Ikeda, K., Nakamura, Y., & Humble, T. S. (2019). Application of Quantum Annealing to Nurse Scheduling Problem. *Scientific Reports*, 9, 12837. DOI: 10.1038/s41598-019-49172-3.
4. Larsson, N., & Kanerot, S. (2025). Quantum Optimization of Physician Scheduling For Maximal Healthcare Capacity. Chalmers University of Technology.
5. QTIS: A QAOA-based Quantum Time Interval Scheduler. *Future Generation Computer Systems*, 183, 108594 (2026). DOI: 10.1016/j.future.2026.108594.
6. Waseem, H. M., Epiphaniou, G., & Maple, C. (2026). Quantum-Assisted Optimization and Security for Trustworthy AI-Driven Healthcare. *IEEE Transactions on Quantum Engineering*, 7, 2100825. DOI: 10.1109/TQE.2026.3677420.